Rendeiro 2020 Chronic Lymphocytic Leukemia — analysis summary

RUN 3 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) fuses the four patient-private CLL clones: per-neighbourhood iLISI 1.0→~5.1 / 7 across a θ×n_pcs sweep (plateau at θ=10; θ=20 over-collapses), one connected kNN component after island removal. Decisive check: the Gast→Crest gradient reproduces WITHIN each clone (Crest↔Gast −0.47…−0.58 in every patient), so integration aligns a shared intra-clonal gradient, not a fabricated continuum; proliferation diffuse (corr≈0).

Gate 2 — module geography → GO (non-branching)

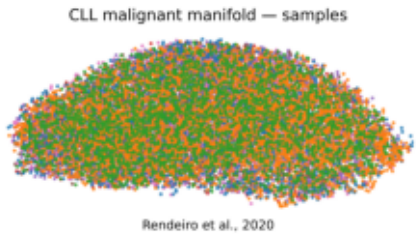
Signatures collapse onto a single (Gast/pSom)→Crest axis: raw per-cell Crest↔Gast r=−0.55, Crest↔pSom r=−0.55, Gast↔pSom r≈+0.05 (no cross-repression — no two competing arms). Crest is an inverse terminal gradient. Principal curve is fold-free (2 tips / 0 forks; self-approach 1.6). Crest is NOT enriched on a Gast arm — no inversion flag.

Gate 3 — hub cross-repression → NA

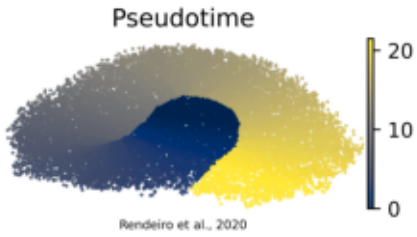
Non-branching topology — no bifurcation, no Gast/pSom hub, so the branching-only cross-repression test does not apply. PPT tree gives only unstable spurious forks (4→6→9 as nodes 100→200) over a blob. Modules notebooks not produced.

STEP FIGURES

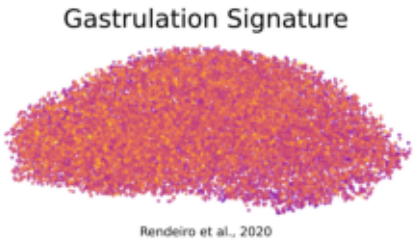
Manifold — samples



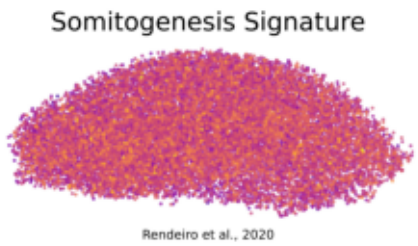
Pseudotime



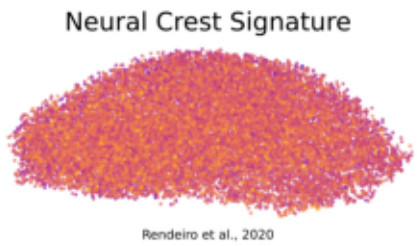
Gastrulation



Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Chronic lymphocytic leukemia (Rendeiro et al., 2020), 10x — 18,314 malignant B-cells over 7 samples / 4 patients.

Verdict.
Pattern reproduces as the NON-BRANCHING case — a single gastrulation/somitogenesis → neural-crest developmental axis; no hub, no bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means (raw) Gast −0.017 / pSom +0.007 / Crest +0.007 — Linear. EMT-TFs ZEB2 (12.1%) + ZEB1 (3.8%); Snail/Twist/FOXC/PRRX silent.

Parameter choices

Min-cell threshold	285 — geom mean of the 1012→80 spillover cliff; keeps 7/11 samples, 4 patients	**Integration**	Harmony θ=10 (patient-private CLL clones), n_pcs=30, spread=1.3
Islands removed	343 cells / 57 detached specks (UMAP 6× radius graph, largest component)	**Geometry**	Non-branching scFates curve (ElPiGraph), Nodes=20; 2 tips / 0 forks, fold-free (SA 1.6)
Root node	curve tip 2 = Gast-high, Crest-low (origin; Crest terminal along pseudotime)	**Module means**	Gast −0.017 / pSom +0.007 / Crest +0.007 — Linear
EMT-TFs	ZEB2 (12.1%) + ZEB1 (3.8%); Snail/Twist/FOXC/PRRX silent		